ALGEBRA 1 · POLYNOMIALS

Area Model Multiplier

Explore & Reflect

Name _____ Date _____ Period _____

Open **Area Model Multiplier** on Mr. K's Classroom: Algebra 1 → Polynomials → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Is $(x + 3)^2$ equal to $x^2 + 9$? Explain.

2. How many different kinds of pieces does the rectangle for $(x + 2)(x + 5)$ have?

2 Explore use the tool and record what you see

Build each product

Record each part of the rectangle, then combine.

Product	x^2 part	x parts	Number part	Simplified
$(x + 2)(x + 3)$				
$(x + 4)(x - 1)$				
$(x + 3)^2$				
$(x + 5)(x - 5)$				
$(2x + 1)(x + 3)$				

3 Write the rule fill in each blank from what you observed

1. The middle term comes from _____ the two x parts.

2. The constant term is the _____ of the two constants.

3. $(x + a)^2 = x^2 +$ _____ $+ a^2$, not $x^2 + a^2$.

4. $(x + a)(x - a) =$ _____, because the two x parts cancel.

4 Check no tool: show your work

1. Multiply $(x + 6)(x + 2)$.

2. Multiply $(x - 3)(x - 7)$.

3. Multiply $(x - 4)^2$.

4. Multiply $(3x + 2)(x - 1)$.

5 Reflect

Draw or describe the rectangle for $(x + 1)(x + 4)$ and explain where each term comes from.